

15. (g) Implementation Plan: B.Tech. in Chemical Engineering

Semester – I							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21LEH101T	Communicative English	2	1	0	3		
21MAB101T	Calculus and Linear Algebra	3	1	0	4		
21PYB101J	Physics: Electromagnetic Theory, Quantum Mechanics, Waves and Optics	3	1	2	5		
21MES102L ¹	Engineering Graphics and Design	0	0	4	2		
21EES101T	Electrical and Electronics Engineering	3	1	0	4		
21CYM101T ¹	Environmental Science*	1	0	0	0		
21PDM101L ¹	Professional Skills and Practices	0	0	2	0		
21LEM101T ¹	Constitution of India	1	0	0	0		
Total Credits					18		
Semester – II							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21LEH102T	Chinese						
21LEH103T	French						
21LEH104T	German						
21LEH105T	Japanese	2	1	0	3		
21LEH106T	Korean						
21LEH107T	Spanish						
21LEH108T	Russian						
21GNH101J	Philosophy of Engineering	1	0	2	2		
21MAB102T	Advanced Calculus and Complex Analysis	3	1	0	4		
21CYB101J	Chemistry	3	1	2	5		
21CHC101J	Physical and Analytical Chemistry	2	0	2	3		
21CSS101J	Programming for Problem Solving	3	0	2	4		
21BTB103T	Biology	2	0	0	2		
21MES101L ¹	Basic Civil and Mechanical Workshop	0	0	4	2		
21PDM102L ¹	General Aptitude*	0	0	2	0		
21GNM101L ¹	Physical and Mental Health using Yoga						
21GNM102L ¹	National Service Scheme	0	0	2	0		
21GNM103L ¹	National Cadet Corps						
21GNM104L ¹	National Sports Organization						
Total Credits					25		
Semester – III							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21MAB201T	Transforms and Boundary Value Problems	3	1	0	4		
21CHC202T	Chemical Process Calculations	3	1	0	4		
21CHC203J	Mechanical Operations	2	0	2	3		
21CHC204J	Chemical Engineering Fluid Mechanics	2	0	2	3		
21PDH209T ¹	Social Engineering	2	0	0	2		
21CHS201T	Introduction to Chemical Engineering	3	0	0	3		
21LEM201T ¹	Professional Ethics	1	0	0	0		
21PDM201L ¹	Verbal Reasoning	0	0	2	0		
21LEM202T ¹	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	2	1	0	3		
Total Credits					22		
Semester – IV							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21MAB202T	Numerical Methods	3	1	0	4		
21CSC206T	Artificial Intelligence	2	1	0	3		
21CHC205J	Heat Transfer	3	0	2	4		
21CHC206T	Chemical Process Technology	3	0	0	3		
E	Professional Elective – I	3	0	0	3		
21DCS201P ¹	Design Thinking and Methodology	1	2	0	3		
21PDM202L ¹	Critical and Creative Thinking Skills	0	0	2	0		
Total Credits					20		
Semester – V							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21CHC301T ²	Chemical Engineering Thermodynamics	3	1	0	4		
21CHC302J	Mass Transfer Applications	3	0	2	4		
21CHS303J	Computational Methods in Chemical Engineering	2	0	2	3		
21CHC304J	Chemical Reaction Engineering	3	0	2	4		
O	Open Elective – I	3	0	0	3		
E	Professional Elective – II	3	0	0	3		
21PDM301L ¹	Analytical and Logical Thinking Skills	0	0	2	0		
21LEM301T ¹	Indian Art Form	1	0	0	0		
21GNP301L ¹	Community Connect	0	0	2	1		
Total Credits					22		
Semester – VI							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21CSS303T	Data Science	2	0	0	2		
21CHC305J	Process Dynamics, Control and Instrumentation	2	0	2	3		
21CHC306T ²	Transport Phenomena	3	0	0	3		
21CHC307J	Process Modeling and Simulation	2	0	2	3		
E	Professional Elective – III	3	0	0	3		
E	Professional Elective – IV	3	0	0	3		
21CHP303T ¹	MOOC	3	0	0	3		
21CHP302L ¹	Project	0	0	6	3		
O	Open Elective – II	3	0	0	3		
21PDM302L ¹	Employability Skills and Practices	0	0	2	0		
21LEM302T ¹	Indian Traditional Knowledge	1	0	0	0		
Total Credits					23		
Semester – VII							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21GNH401T	Behavioral Psychology	2	1	0	3		
21CHC401J ²	Process Equipment Design and Drawing	2	0	2	3		
21CHC402T	Process Economics and Project Management	3	0	0	3		
E	Professional Elective – V	3	0	0	3		
E	Professional Elective – VI	3	0	0	3		
O	Open Elective – III	3	0	0	3		
Total Credits					18		
Semester – VIII							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21CHP401L	Major Project	0	0	30	15		
21CHP402L	Major Project	0	0	20	10		
21CHP403L	Internship#	0	0	10	5		
Total Credits					15		

#Students have to register either 21CHP401L or 21CHP402L and 21CHP403L both in eighth semester